T4 (refuted): the slope-3 eigenvalue of Π_q is not rational over $\mathbb{Q}(q)$

Clio

2026-04-29

1 The conjecture and what is wrong with it

For each shape λ in the three "atom-cube" shapes

$$\big\{\,(4,2) \text{ at } n=6,\ (4,3) \text{ at } n=7,\ (5,2) \text{ at } n=7\,\big\},$$

the characteristic polynomial $g_\lambda(t) \in \mathbb{Z}[q,t]$ of Π_q acting on the irreducible Specht module V_λ has been computed (Theorems T1, T2, T3 of 2026-04-27/28). The Newton polygon of g_λ with respect to the $\mathfrak{a}$-adic valuation, $\mathfrak{a} := q^2 + 4q + 1$, has the following shape:

$$\begin{aligned}
&(4,2)/n=6 : \text{slopes } -2,-1,0, \text{ each of length 1;}\\
&(4,3)/n=7 : \text{slope } -3 \text{ (length 1) and slope 0 (length 3);}\\
&(5,2)/n=7 : \text{slope } -3 \text{ (length 1) and slope 0 (length 4).}
\end{aligned}$$

The (now-refuted) conjecture proposed:

> **Conjecture T4.** For each shape λ, the unique eigenvalue r_λ of $\Pi_q|V_\lambda$ with positive $\mathfrak{a}$-adic valuation lies in $\mathbb{Q}(q)$ and admits a closed form
>
> $$r_\lambda(q) = q^{a_\lambda}(1+q)^{b_\lambda}\mathfrak{a}(q)^{c_\lambda}\,P_\lambda(q),$$
>
> with $P_\lambda \in \mathbb{Z}[q]$ palindromic-positive.

The justification offered was: *the Newton polygon has a slope $-c_\lambda$ segment of length 1; therefore g_λ admits a linear factor g_1 over $\mathbb{Q}(q)$, so its root is rational.*

This justification is wrong. The Newton-polygon factorisation

$$g_\lambda(t) \;=\; g_1(t)\cdot g_0(t)$$

takes place *over the completion* $\mathbb{Q}(q)^{\mathfrak{a}}$, not over $\mathbb{Q}(q)$ itself. The factor g_1 is linear over the completion, so its root lies in $\mathbb{Q}(q)^{\mathfrak{a}}$, but *a priori* the root may not lie in the global field $\mathbb{Q}(q)$. The conjecture conflates "length-1 segment" with "$\mathbb{Q}(q)$-linear factor"; only the former is automatic.

This note proves the conjecture is false in the strongest possible way: $g_\lambda(t)$ is in fact *irreducible* over $\mathbb{Q}(q)$ for every one of the three shapes, so *no* eigenvalue lies in $\mathbb{Q}(q)$.

2 The theorem

Theorem 1. *For each of the three shapes $\lambda \in \{(4,2)/n = 6,\ (4,3)/n = 7,\ (5,2)/n = 7\}$, the characteristic polynomial $g_\lambda(t) \in \mathbb{Z}[q,t]$ of Π_q acting on the Specht module V_λ is* irreducible *over $\mathbb{Q}(q)$.*

Corollary 2. *None of the eigenvalues of Π_q on V_λ lies in $\mathbb{Q}(q)$. In particular, the conjectured rational closed form $r_\lambda(q)$ does not exist for any of the three atom-cube shapes.*

The corollary is immediate: each eigenvalue of $\Pi_q|V_\lambda$ is a root of $g_\lambda(t)$. Were any such root in $\mathbb{Q}(q)$, the factor $(t-r) \in \mathbb{Q}(q)[t]$ would be a linear factor of g_λ, contradicting Theorem 1.

3 Proof of Theorem 1

The key ingredient is the standard specialization criterion.

Lemma 3. *Let $g(q,t) \in \mathbb{Z}[q,t]$ be monic of positive degree in t. If there exists $q_0 \in \mathbb{Z}$ such that the specialization $g(q_0,t) \in \mathbb{Z}[t]$ is irreducible over $\mathbb{Q}$, then $g(q,t)$ is irreducible over $\mathbb{Q}(q)$.*

Proof. Suppose, for contradiction, that g factors over $\mathbb{Q}(q)$ as $g = f_1 \cdot f_2$ with $f_1, f_2 \in \mathbb{Q}(q)[t]$ both of positive t-degree. Since g is monic in t, by absorbing units we may take f_1, f_2 each monic in t. By Gauss's lemma applied to $\mathbb{Z}[q,t] = \mathbb{Z}[q][t]$, the monic factors f_1, f_2 in fact lie in $\mathbb{Z}[q][t]$. Specializing at $q = q_0$, we obtain

$$g(q_0,t) \;=\; f_1(q_0,t)\, f_2(q_0,t) \in \mathbb{Z}[t],$$

where each factor is monic in t of the same positive degree as f_i, hence a non-trivial factorization over $\mathbb{Z}$ (and so over $\mathbb{Q}$). This contradicts the assumed irreducibility of $g(q_0,t)$ over $\mathbb{Q}$. □

Proof of Theorem 1. By Lemma 3, it suffices to exhibit, for each shape λ, a single value $q_0 \in \mathbb{Z}$ at which $g_\lambda(q_0,t) \in \mathbb{Z}[t]$ is irreducible over $\mathbb{Q}$.

We take $q_0 = 2$ throughout. The polynomial g_λ is determined from the elementary symmetric functions $\sigma_1, \ldots, \sigma_r$ of the eigenvalues (where r is the rank of the Hecke-invariant subspace, the Specht-internal rank that selects the dimension of the image of Π_q). These $\sigma_k(q)$ were computed exactly via Lagrange interpolation in T1, T2, T3 (*cf.* the proof documents `2026-04-27-newton-polygon-proof.tex` and `2026-04-28-newton-polygon-n7-concentrated.tex`).

Shape $(4,2)$ at $n = 6$ ($r = 3$). Specializing at $q_0 = 2$:

$$g_{(4,2)}(2,t) = t^3 - 2{,}479{,}572\,t^2 + 726{,}098{,}626{,}656\,t - 11{,}766{,}474{,}745{,}339{,}392.$$

Direct factorisation of this cubic over $\mathbb{Q}$ (or, equivalently, over $\mathbb{Z}$ by Gauss) yields the polynomial as a single irreducible factor. The rational root theorem rules out rational roots since none of the divisors of 11,766,474,745,339,392 is a root, and a degree-3 polynomial over $\mathbb{Q}$ is reducible iff it has a rational root. Verified computationally via SymPy's `Poly(·, t, domain="QQ").factor_list()`.

Shape $(4,3)$ at $n = 7$ ($r = 4$). Specializing at $q_0 = 2$:

$$\begin{aligned} g_{(4,3)}(2,t) = t^4 &- 709{,}843{,}176\,t^3 + 63{,}282{,}432{,}370{,}649{,}472\,t^2 \\ &- 469{,}939{,}687{,}270{,}795{,}269{,}181{,}440\,t \\ &+ 191{,}646{,}477{,}786{,}342{,}535{,}098{,}958{,}872{,}576. \end{aligned}$$

SymPy's `factor_list` over $\mathbb{Q}$ returns a single irreducible factor.

Shape $(5,2)$ **at** $n=7$ **(**$r=5$**).** Specializing at $q_0=2$:

$$\begin{aligned} g_{(5,2)}(2,t) = t^5 &- 2{,}412{,}957{,}924\,t^4 + 998{,}432{,}827{,}755{,}810{,}336\,t^3 \\ &- 70{,}028{,}605{,}693{,}778{,}159{,}853{,}609{,}984\,t^2 \\ &+ 1{,}312{,}296{,}401{,}180{,}928{,}340{,}893{,}912{,}771{,}502{,}080\,t \\ &- 2{,}004{,}689{,}847{,}756{,}036{,}486{,}677{,}534{,}364{,}715{,}630{,}460{,}928. \end{aligned}$$

SymPy's `factor_list` over $\mathbb{Q}$ returns a single irreducible factor.

In each case the specialization $g_\lambda(2,t)$ is irreducible over $\mathbb{Q}$, so by Lemma 3 the polynomial $g_\lambda(q,t)$ is irreducible over $\mathbb{Q}(q)$. □

Remark (Cross-check). We further verified irreducibility of $g_\lambda(q_0,t)$ at $q_0=3$ for all three shapes and at $q_0=5$ for $(4,2)$. Both factorisations again returned a single irreducible factor, in line with the fact that the irreducibility statement at one specialization already implies irreducibility globally. Independent SymPy factorisations over $\mathbb{Q}(q)[t]$ directly (using `Poly(·, t, domain=`$\mathbb{Q}(q)$`).factor_list()`) confirm the same: each g_λ remains a single factor of full degree.

Remark (Galois group). Theorem 1 shows the Galois group $\mathrm{Gal}(L_\lambda/\mathbb{Q}(q))$ of the splitting field L_λ acts transitively on the r_λ eigenvalues. By the Newton-polygon information, the decomposition group at the place $\mathfrak{a}$ has orbits of sizes $(1,1,1)$, $(1,3)$, $(1,4)$ on the eigenvalues for the three shapes respectively, since the Newton-polygon factorisation over the completion $\mathbb{Q}(q)^{\mathfrak{a}}$ has factors of those t-degrees. Determining the full Galois groups is not pursued here; the relevant computational handle is the discriminant of g_λ (easily extracted from the explicit polynomial).

4 Where the original argument breaks down

The error in the original justification of T4 is the conflation of two distinct factorisations:

- **Newton-polygon factorisation over** $\mathbb{Q}(q)^{\mathfrak{a}}$**.** Over the $\mathfrak{a}$-adic completion of $\mathbb{Q}(q)$, the polynomial g_λ splits as
$$g_\lambda(t) \;=\; g_1(t)\cdot g_0(t),$$
where g_1 has all roots of $\mathfrak{a}$-adic valuation $c_\lambda=3$ (resp. 2, 1, 0 for the three slopes of $(4,2)$), g_0 has all roots of valuation 0, and $\deg g_1$ equals the length of the corresponding slope. This is automatic from the polygon and is used in the proofs of T1, T2, T3.

- **Polynomial factorisation over** $\mathbb{Q}(q)$**.** This is the question of whether $g_\lambda \in \mathbb{Q}(q)[t]$ factors. Theorem 1 says it does not.

These two factorisations agree exactly when $\mathbb{Q}(q)$ is *Henselian* at the place $\mathfrak{a}$, equivalently when $\mathbb{Q}(q)$ equals its $\mathfrak{a}$-adic completion — which is far from true. The atom $\mathfrak{a}=q^2+4q+1$ is irreducible of degree 2 in $\mathbb{Q}[q]$ (its discriminant is 12, not a square), so the residue field $\mathbb{Q}[q]/(\mathfrak{a}) \cong \mathbb{Q}(\sqrt{3})$ is a non-trivial extension of $\mathbb{Q}$. The completion $\mathbb{Q}(q)^{\mathfrak{a}}$ contains, but strictly, $\mathbb{Q}(q)$.

5 What survives

Proposition 4 (Atom-adic existence and uniqueness, verified)**.** *For each shape λ, there is a* unique *root r_λ of g_λ in the completion $\mathbb{Q}(q)^{\mathfrak{a}}$ with positive $\mathfrak{a}$-adic valuation, and that valuation equals c_λ (the highest atom-exponent dictated by the polygon: 2, 3, 3 for $(4,2)$, $(4,3)$, $(5,2)$ respectively). This root lies in $\mathbb{Q}(q)^{\mathfrak{a}}$ but* not *in $\mathbb{Q}(q)$.*

This was the genuine content extractable from the Newton polygon argument; only the *rationality* claim was unwarranted.

Proposition 5 (Leading coefficient in residue field)**.** *Write $r_\lambda = \mathfrak{a}^{c_\lambda} \cdot u_\lambda$ with u_λ a unit in the local ring of $\mathbb{Q}(q)^{\mathfrak{a}}$. Reducing modulo $\mathfrak{a}$ gives a leading coefficient $\overline{u}_\lambda \in \mathbb{Q}(\alpha)$, where α is a root of $\mathfrak{a}$, i.e. $\alpha = -2 + \sqrt{3}$.*

A direct computation (from the leading Hensel approximation $r_\lambda \equiv \sigma_r/\sigma_{r-1} \pmod{\mathfrak{a}^{c_\lambda+1}}$, which is valid since $\mathrm{ord}_{\mathfrak{a}}(g(\sigma_r/\sigma_{r-1})) = c_\lambda + 1$ in each case) yields:

$$\begin{aligned}
\overline{u}_{(4,2)} &= \tfrac{-850720-290912\,\alpha}{68361} \in \mathbb{Q}(\alpha),\\
\overline{u}_{(4,3)} &= \tfrac{78240950912+28791367552\,\alpha}{7949071559} \in \mathbb{Q}(\alpha),\\
\overline{u}_{(5,2)} &= \tfrac{-2281470720+1742825728\,\alpha}{1128105863} \in \mathbb{Q}(\alpha).
\end{aligned}$$

The denominators arise from rationalising $\overline{P_{r-1}(\alpha)}^{-1}$, where P_{r-1} is the remainder polynomial appearing in $\sigma_{r-1} = q^a(1+q)^b \cdot P_{r-1}(q)$ with P_{r-1} coprime to $\mathfrak{a}$.

These $\overline{u}_\lambda$ are not in $\mathbb{Q}$ (they have nonzero α-coefficient), confirming once more that r_λ is not $\mathbb{Q}(q)$-rational.

The cross-shape pattern from $A_9 = q^6 + 12q^5 + 52q^4 + 88q^3 + 52q^2 + 12q + 1$ (which appears as the palindromic core of σ_r for *both* $(4,3)$ and $(5,2)$ at $n = 7$) does *not* translate into a clean cross-shape relation between $\overline{u}_{(4,3)}$ and $\overline{u}_{(5,2)}$: the denominators 7,949,071,559 and 1,128,105,863 have no obvious common structure.

6 Computational verification

Code. `~/projects/scratch/2026-04-29-rational-atom-eigenvalue/`

- `compute_eigenvalue.py` — interpolates $\sigma_k(q)$, forms $g_\lambda(t)$, factors over $\mathbb{Q}(q)[t]$ via SymPy.
- `specialize_check.py` — specializes $g_\lambda(q_0, t)$ at $q_0 \in \{2, 3, 5\}$, factors over $\mathbb{Q}[t]$.
- `check_extensions.py` — factors over $\mathbb{Q}(\sqrt{3})(q)[t]$ (residue-field extension).
- `leading_unit.py` — extracts the leading residue-field coefficient $\overline{u}_\lambda \in \mathbb{Q}(\alpha)$.
- `verify_irreducibility.py` — Newton iteration showing σ_r/σ_{r-1} is not a root.

Verifications run.

1. Each $\sigma_k(q) \in \mathbb{Z}[q]$ Lagrange-interpolated from 89/50/110 integer evaluations (overdetermined relative to the degree bound; cross-checked).
2. $g_\lambda(t)$ over $\mathbb{Q}(q)[t]$: `Poly(·, t, domain=QQ(q)).factor_list()` returns a single factor of full t-degree for each shape.
3. $g_\lambda(q_0, t)$ over $\mathbb{Q}[t]$ for $q_0 = 2, 3, 5$: irreducible in every case.
4. $g_\lambda(t)$ over $\mathbb{Q}(\sqrt{3})(q)[t]$: still irreducible. So the slope-c_λ root does not lie even in the unramified extension corresponding to the residue field $\mathbb{Q}(\alpha)$.

7 Status

T4 (the rational atom-eigenvalue conjecture) is **refuted**.

- The eigenvalues of Π_q on the three atom-cube Specht modules are not individually in $\mathbb{Q}(q)$.
- They generate a degree-r extension of $\mathbb{Q}(q)$ (namely $\mathbb{Q}(q)[t]/(g_\lambda)$).
- The original Newton-polygon argument was misapplied: the polygon factorisation lives over the completion, not the global field.
- What is preserved: the symmetric functions σ_k (the Eigenvalue Rosetta Stone factorisation $\sigma_k = q^a(1+q)^b\mathfrak{a}^c P_k(q)$ with P_k palindromic) — these are $\mathbb{Q}(q)$-rational by construction. The atom-exponent grading of the eigenvalue *multiset* (T1, T2, T3) is unchanged.

The σ_k-G1 program (palindromic-positivity at level σ_k) is unaffected: it is a statement about the symmetric functions, not about individual eigenvalues. So T4 was a *stretch goal* in the original PROVE.md ("what is the closed form of the atom-cube eigenvalue?"), and the answer is now: there is no $\mathbb{Q}(q)$-closed form, because no such closed form exists.

Open directions.

- **Galois group.** Compute $\mathrm{disc}(g_\lambda)$ and determine $\mathrm{Gal}(g_\lambda/\mathbb{Q}(q))$. For $(4,2)$ this is a transitive subgroup of S_3 — either S_3 or $A_3 = \mathbb{Z}/3$. For $(4,3)$ it is a transitive subgroup of S_4. For $(5,2)$ it is a transitive subgroup of S_5.
- **Higher-order Hensel coefficients.** The expansion $r_\lambda = \mathfrak{a}^{c_\lambda}\overline{u}_\lambda + \mathfrak{a}^{c_\lambda+1}\overline{u}_\lambda^{(1)} + \cdots$ has each coefficient in $\mathbb{Q}(\alpha)$. Whether these coefficients have any pattern (e.g. all in $\mathbb{Q}$ from some point on, or following a recursion) is a natural follow-up.
- **Why g is irreducible.** What categorical / Hecke structural reason forces the eigenvalues of Π_q on V_λ to be Galois-conjugate over $\mathbb{Q}(q)$ for these three shapes? This is the genuinely interesting question that emerges from the refutation.